□ Data Analyst Assignment

📘 Objective

Evaluate the candidate's ability to:

- Clean and analyze datasets using **Python** and **SQL**
- Create visual dashboards using **Power BI**
- Derive insights and communicate findings effectively

□ Part 1: SQL – Data Exploration & Analysis

Dataset:

Provide a sample dataset (e.g., `sales_data.csv`) containing:

order_id	customer_id	product_category	product_name	quantity	price	order_date	region
1001	C001	Electronics	Laptop	1	55000	2023-01-10	South
1002	C002	Furniture	Chair	4	1500	2023-01-12	North
...	...	...	...	...	...	...	...

Tasks:

1. Write an SQL query to find the **top 5 best-selling products** by revenue.
2. Calculate **monthly total sales and total quantity sold**.
3. Find the **average order value** by region.
4. Identify customers who have placed **more than 3 orders**.
5. Get the **percentage contribution of each category** to total revenue.

🐍 Part 2: Python – Data Cleaning & Analysis

Dataset:

Use the same `sales_data.csv`.

Tasks:

1. **Load the data** using pandas and check for missing values.

2. Handle missing data (e.g., fill, drop, or impute) and remove duplicates.
3. Create a new column `total_amount = quantity * price`.
4. Find:
   - Total revenue generated
   - Most sold product category
   - Region with highest sales
5. Plot:
   - Monthly sales trend (line chart)
   - Category-wise revenue (bar chart)
6. Export the cleaned dataset to a new CSV (`cleaned_sales_data.csv`).

📊 Part 3: Power BI – Dashboard Creation

Dataset:

Use the `cleaned_sales_data.csv` from the Python section.

Tasks:

Create a Power BI dashboard with the following:

1. **KPIs:** Total Revenue, Total Orders, Average Order Value
2. **Charts:**
   - Sales by Region (map or bar chart)
   - Sales Trend by Month (line chart)
   - Category-wise Revenue (pie or bar chart)
   - Top 5 Products by Revenue
3. **Filters:** Region, Product Category, Date Range
4. **Insight Section:** Write 3–5 bullet points summarizing insights (e.g., “Electronics category contributes 40% of total revenue”).

☐ Bonus (Optional)

- Build a simple **predictive model in Python** to forecast next month’s sales using linear regression.
- Add a **Power BI page** for “Forecasted Sales”.

🎁 Submission Format

- Folder containing:
  - `SQL_queries.sql`
  - `data_analysis.ipynb`

  - `cleaned_sales_data.csv`
  - `PowerBI_dashboard.pbix`
  - `Insights_Report.pdf` (1–2 pages summary of key findings)
- Deadline: (Till Diwali vaction.)

✅ Evaluation Criteria

Criteria	Weight
SQL query accuracy & logic	25%
Python data cleaning & analysis	30%
Power BI visualization quality	30%
Insights & presentation clarity	15%
